Name

Mathematics Advanced Year 12

Quadratic and Cubic Functions

Book 1

Version: 261021

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Contents

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

Quadratic and cubic functions

- Identify the x-intercepts of a parabola whose quadratic function is expressed in factored form
- Use the discriminant to determine the number of x-intercepts on a parabola and justify its position in relation to the x-axis
- Show by completing the square on the general quadratic $y = ax^2 + bx + c$ that the axis of symmetry is $x = -\frac{b}{2a}$ where a, b and c are constants
- Identify the axis of symmetry and vertex of a parabola by completing the square on its quadratic function
- Choose and apply appropriate techniques to graph a parabola of the form $y = ax^2 + bx + c$ by identifying its x-intercepts if they exist, its y-intercept, its axis of symmetry using $x = -\frac{b}{2a}$ and its vertex
- Find the equation of a parabola given sufficient graphical features
- Use the fact that two quadratic functions are equal for all values of x if and only if the corresponding coefficients are equal to solve related problems
- Recognise that solving $f(x) = k$ for some constant k corresponds to finding the x-coordinate(s) of the intersection of the graphs $y = f(x)$ and $y = k$
- Solve problems by finding the solution to simultaneous equations involving a linear and a quadratic function, or two quadratic functions, both algebraically and graphically
- Solve quadratic inequalities
- Recognise and graph cubic functions of the form $f(x) = kx^3$ and $f(x) = k(x - a)(x - b)(x - c)$, where a, b, c and k are constants and $k \neq 0$

Sketching Parabolas by Factorisation

ⓘ General Form of a Parabola

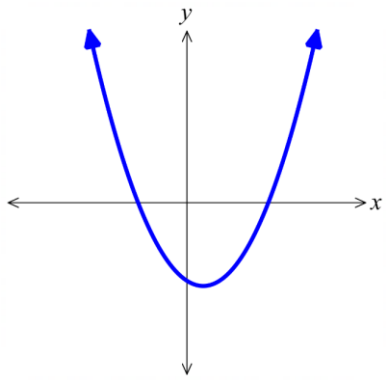
The **general form** of a parabola tells us the concavity and y-intercept.

$$y = ax^2 + bx + c$$

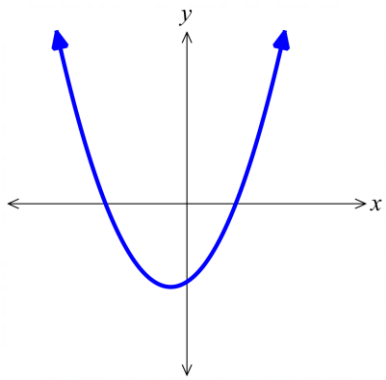
Concavity (a) — y-intercept (c)

🔍 Identify y-intercept and concavity from the equation of a parabola

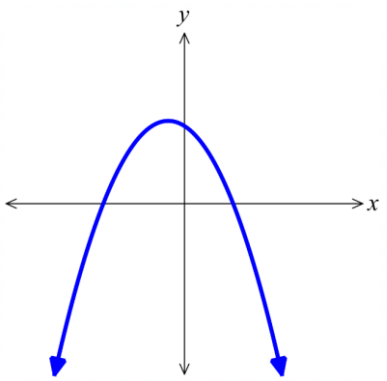
a $y = x^2 - 2x - 15$



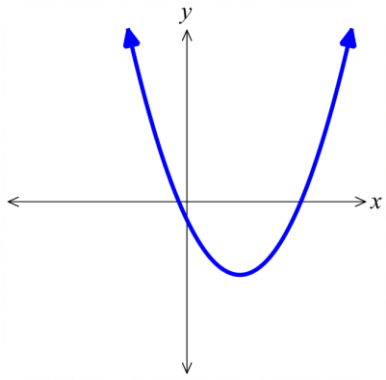
b $y = x^2 + 2x - 15$



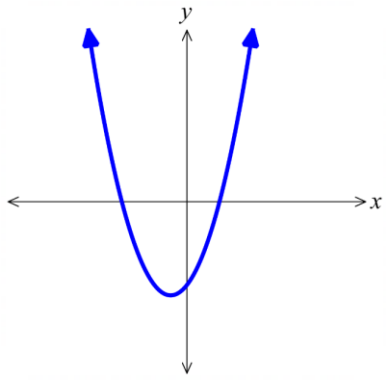
c $y = -x^2 - 2x + 15$



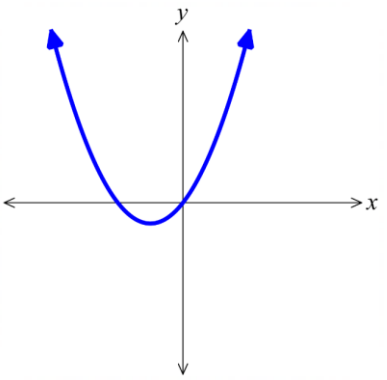
d $y = x^2 - \frac{13}{2}x - \frac{7}{2}$



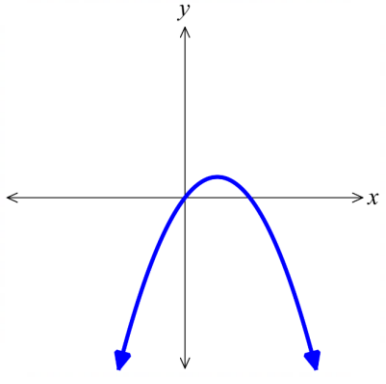
e $y = 2x^2 + 4x - 16$



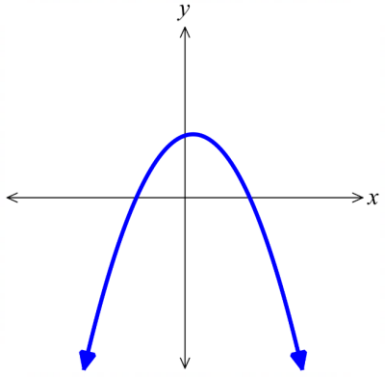
f $y = x^2 + 4x$



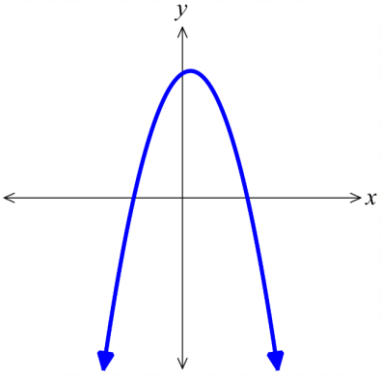
g $y = -x^2 + 4x$



h $y = -x^2 + x + 12$



i $y = -2x^2 + 2x + 24$



ⓘ Intercept Form of a Parabola

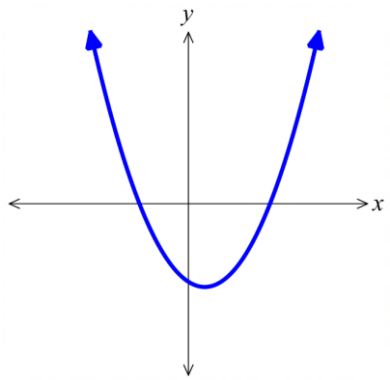
The **intercept form** of a parabola reveals its x-intercepts, since setting $y = 0$ allows the solutions to be found using the null factor law.

$$y = a(x - \alpha)(x - \beta)$$

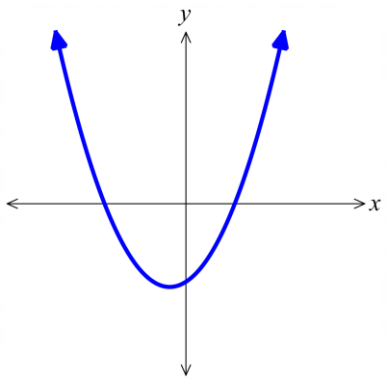
Concavity — x-intercepts

Identify x-intercepts from the equation of a parabola

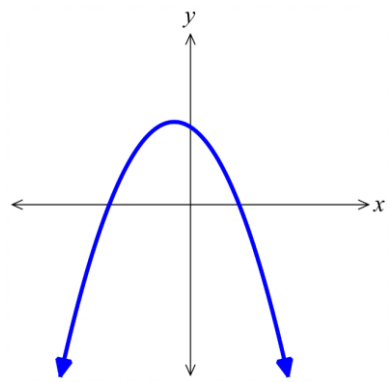
a $y = (x + 3)(x - 5)$



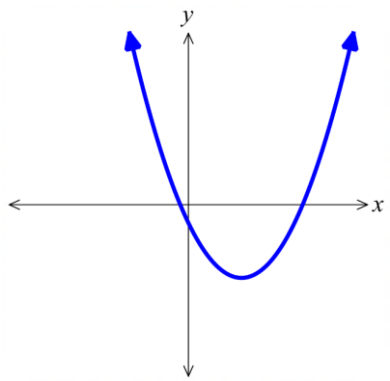
b $y = (x - 3)(x + 5)$



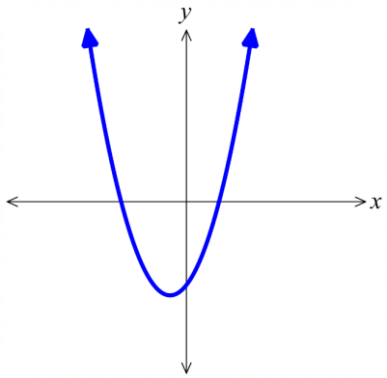
c $y = -(x - 3)(x + 5)$



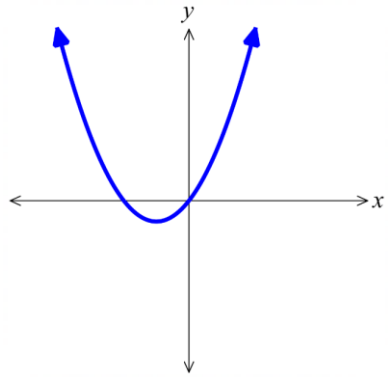
d $y = \left(x + \frac{1}{2}\right)(x - 7)$



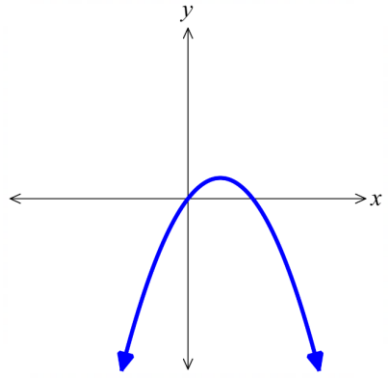
e $y = (2x + 8)(x - 2)$



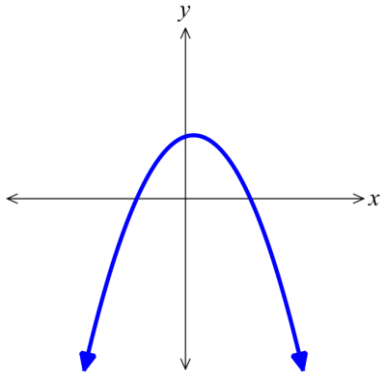
f $y = x(x + 4)$



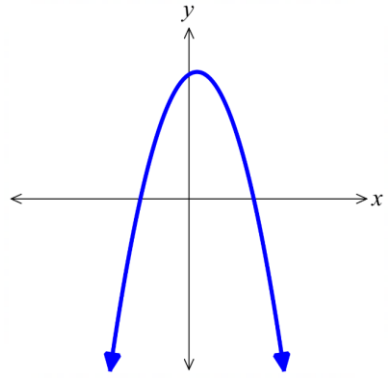
g $y = x(4 - x)$



h $y = (x + 3)(4 - x)$



i $y = (2x + 6)(4 - x)$



Sketch Parabolas by Factorisation

0. Factorise $y = ax^2 + bx + c$ to $y = a(x - \alpha)(x - \beta)$.
1. Determine the **concavity**. $a > 0$, concave up U $a < 0$, concave down ∩.
2. Find the **y-intercept**, $x = 0$.
3. Find the **x-intercepts**, if any. $y = 0$.
4. Determine the **turning point**.
   Use the mean of the x-intercepts to find the x-coordinate,
   then substitute into the equation to find the y-coordinate.

Example Sketch parabolas from factored form

Sketch $y = x^2 - 6x + 5$

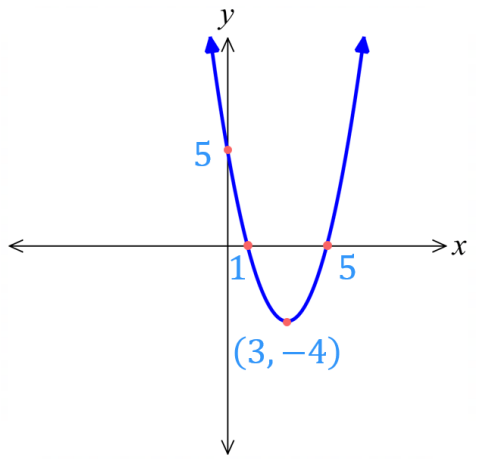
0. Factorise: $y = (x - 5)(x - 1)$

1. Concavity: $a > 0$, concave up

2. y-int: Let $x = 0$
   $y = 5$

3. x-int: Let $y = 0$
   $0 = (x - 5)(x - 1)$
   $x = 5$ or 1

4. TP: Avg of x-intercepts: $\frac{1+5}{2} = 3$
   Sub $x = 3$ into equation
   $y = 3^2 - 6(3) + 5$
   $y = -4$



Guided Practice

Sketch $y = x^2 + 4x - 12$

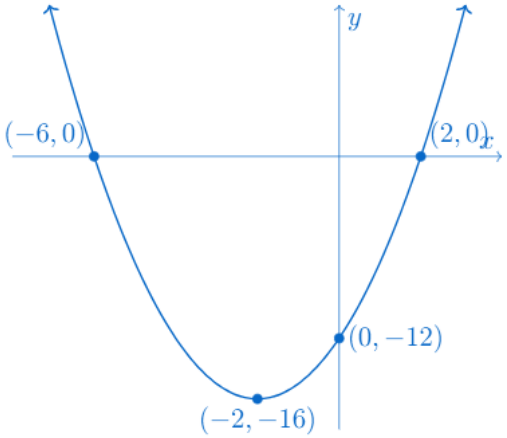
0. Factorise:

1. Concavity:

2. y-int:

3. x-int:

4. TP:



Guided Practice

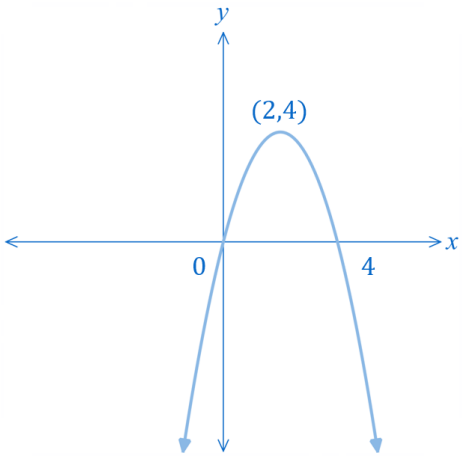
a $y = -x^2 + 4x$

0. Factorise:

1. Concavity:

2. y-int:

3. x-int:



4. Vertex:

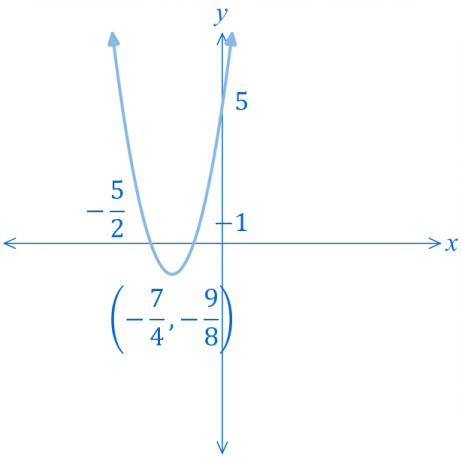
b $y = 2x^2 + 7x + 5$

0. Factorise:

1. Concavity:

2. y-int:

3. x-int:



4. Vertex:

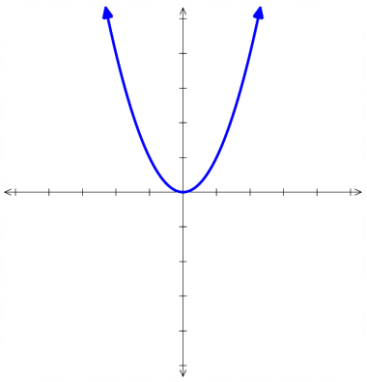
Parabola Transformations

ⓘ Parabola Transformations

For a parabola $y = k(x - b)^2 + c$

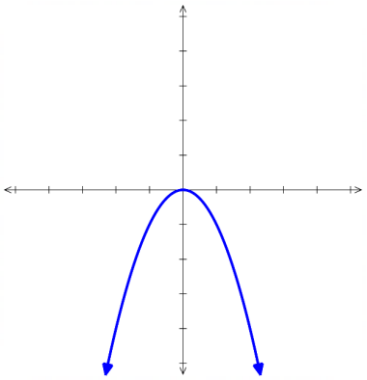
$k > 0$, the parabola is concave ……, with a …… turning point.

$y = 1x^2$



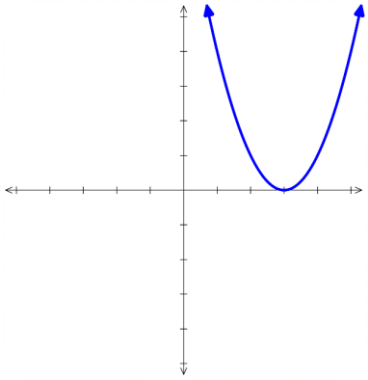
$k < 0$, the parabola is concave ……, with a …… turning point.

$y = -1x^2$



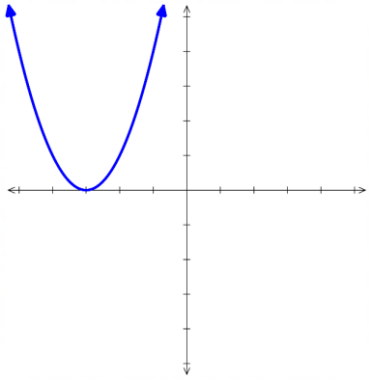
$b > 0$, the parabola is shifted ……

$y = (x - 3)^2$



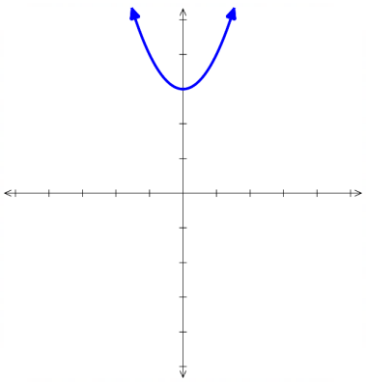
$b < 0$, the parabola is shifted ……

$y = (x + 3)^2$



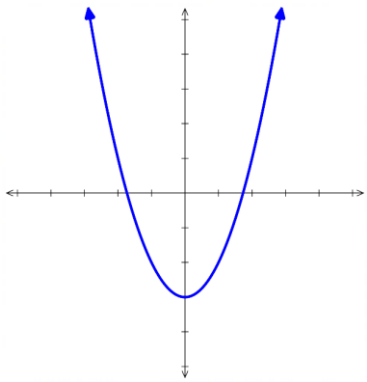
$c > 0$, the parabola is shifted ……

$y = x^2 + 3$



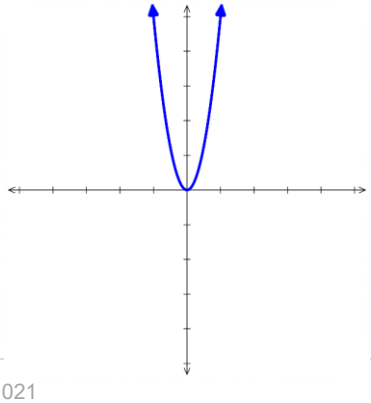
$c < 0$, the parabola is shifted ……

$y = x^2 - 3$



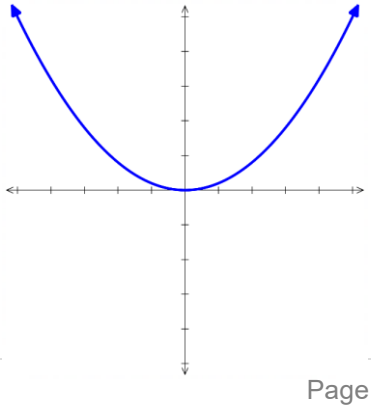
$|k| > 1$, the parabola gets ……

$y = 5x^2$



$0 < |k| < 1$, the parabola gets ……

$y = \frac{1}{5}x^2$



ⓘ Vertex Form of a Parabola

The **vertex form** of a parabola reveals its **turning point** because it shows the graph's horizontal and vertical shifts.

$$y = a(x-h)^2 + k$$

↑ Concavity ↑ turning point (h,k) ↑

🔍 **Identify** turning point of a parabola

Equation	Turning Point (Vertex)
$y = 3(x-4)^2 + 2$	$(4,2)$
$y = 3(x-4)^2 - 2$	$(4,-2)$
$y = -3(x-4)^2 - 2$	$(4,-2)$
$y = -3(x+4)^2 - 2$	$(-4,-2)$
$y = -3(x+4)^2 + 2$	$(-4,2)$
$y = \frac{1}{3}(x+4)^2 + 2$	$(-4,2)$
$y = \frac{(x+4)^2}{3} + 2$	$(-4,2)$
$y = \frac{(x+4)^2 + 2}{3}$	$\left(-4,\frac{2}{3}\right)$
$y = (x+4)^2 + 2$	$(-4,2)$
$y = (x+4)^2$	$(-4,0)$
$y = x^2$	$(0,0)$
$y = x^2 + 5$	$(0,5)$
$y = \frac{x^2}{5}$	$(0,0)$
$y = -9(x+1)^2 - 5$	$(-1,-5)$

Sketching Parabolas from Vertex Form

Sketch Parabolas from Vertex Form

To sketch $y = a(x-h)^2 + k$:

1. Determine the **concavity**.

   $a > 0$, concave up U $a < 0$ concave down ∩.

2. Find the **y-intercept**. Substitute $x = 0$.
3. Find the **x-intercepts**, if any. Substitute $y = 0$.
4. Determine the **vertex**. Using (h, k).

Example Sketch a parabola from vertex form

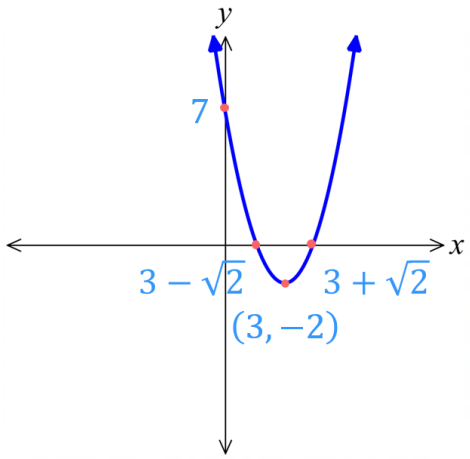
Sketch $y = (x-3)^2 - 2$

1. Concavity: $a > 0$, concave up.

2. y-int: Let $x = 0$
   $y = (0-3)^2 - 2 = 7$

3. x-int: Let $y = 0$
   $0 = (x-3)^2 - 2$
   $2 = (x-3)^2$
   $\pm\sqrt{2} = x - 3$
   $3 \pm \sqrt{2} = x$
   x-ints at $3+\sqrt{2}$ and $3-\sqrt{2}$

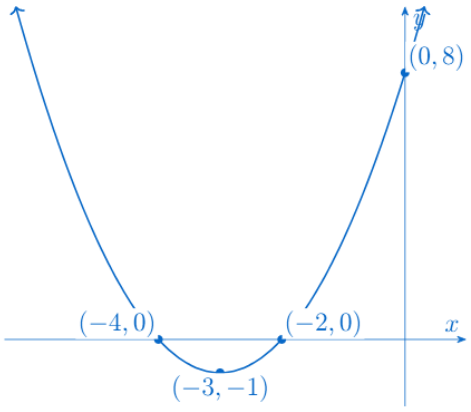
4. Vertex: $(3, -2)$



Guided Practice

Sketch $y = (x+3)^2 - 1$

1. Concavity:
2. y-int:
3. x-int:
4. Vertex:



Guided Practice

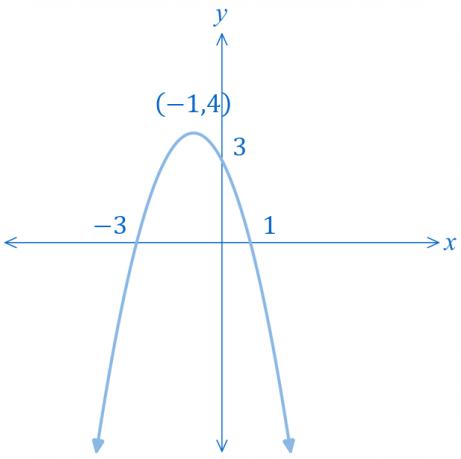
a $y = -(x+1)^2 + 4$

1. Concavity:

2. y-int:

3. x-int:

4. Vertex:



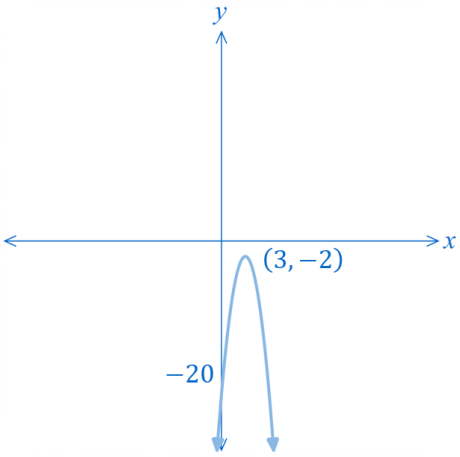
b $y = -2(x-3)^2 - 2$

1. Concavity:

2. y-int:

3. x-int:

4. Vertex:



Sketching Parabolas by Completing the Square

ⓘ Sketch Parabolas by Completing the Square

- When factorising by PSF is not possible, **complete the square** to write a quadratic in vertex form.

0. Write in vertex form: halve the coefficient of x, square it, then add and subtract this value.
1. Concavity
2. y-int
3. x-int
4. Vertex

Example Sketch parabolas by completing the square

Sketch $y = x^2 - 4x - 5$

0. Vertex form by CTS

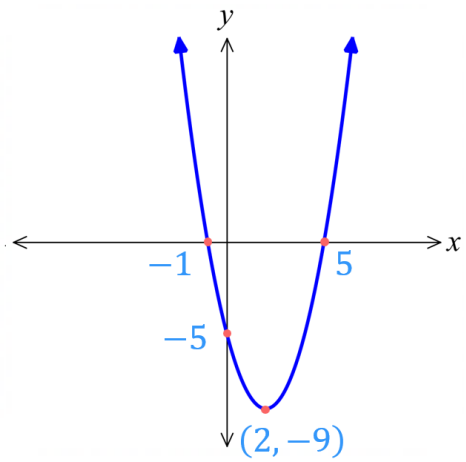
$$y = x^2 - 4x + 4 - 4 - 5$$
$$y = (x^2 - 4x + 4) - 4 - 5$$
$$y = (x - 2)^2 - 9$$

1. Concavity $a > 0$, concave up.

2. y-int: Let $x = 0$,
$y = (0 - 2)^2 - 9 = -5$

3. x-int: Let $y = 0$
$(x - 2)^2 - 9 = 0$
$(x - 2)^2 = 9$
$x - 2 = \pm 3$
$x = 2 \pm 3$
$x = 5$ or -1

4. Vertex: $(2, -9)$



Sketch $y = x^2 + 6x + 15$

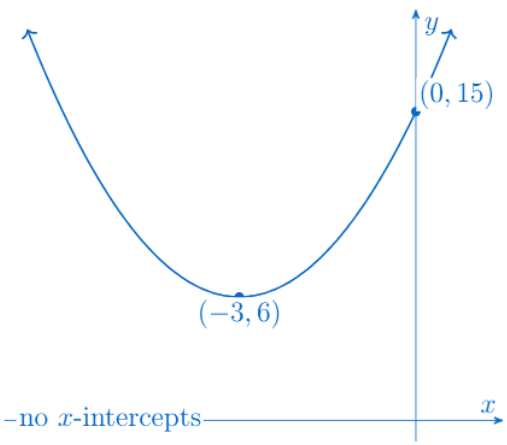
0. Vertex form by CTS

1. Concavity

2. y-int:

3. x-int:

4. Vertex:



Guided Practice

a $y = x^2 - 4x - 1$

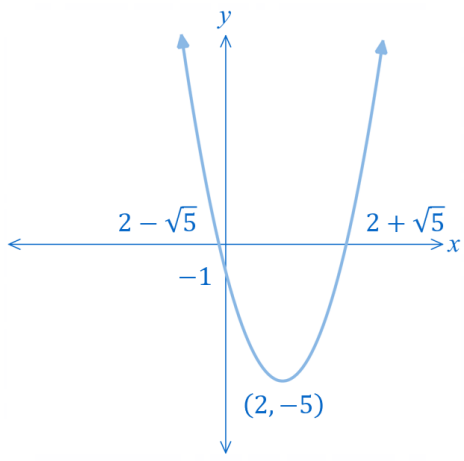
0. Vertex form by CTS

1. Concavity

2. y-int:

3. x-int:

4. Vertex:



b $y = x^2 - 8x - 11$

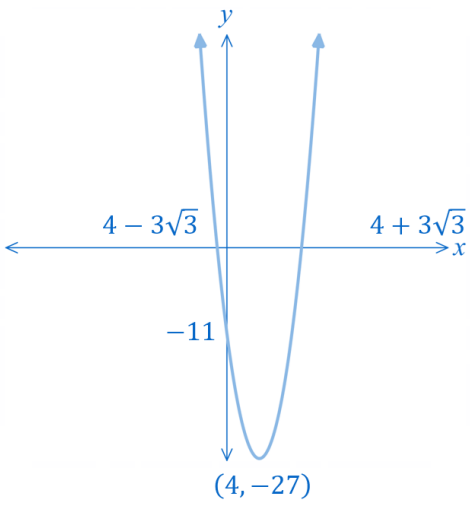
0. Vertex form by CTS

1. Concavity

2. y-int:

3. x-int:

4. Vertex:

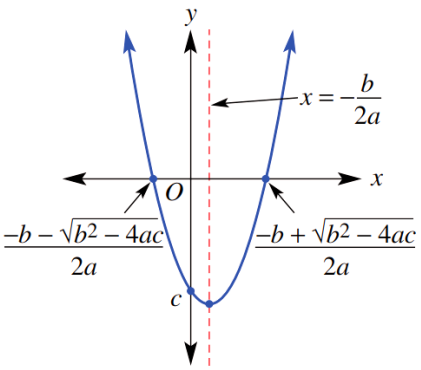


Sketching Parabolas by Quadratic Formula

ⓘ Sketching Parabolas by Quadratic Formula

Instead of completing the square, we can use the quadratic formula to sketch instead.

1. Concavity.
2. y-int.
3. x-ints using $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$
4. Vertex using $x = -\frac{b}{2a}$,
   then substitute value of x into the equation to find y.



Example Sketch parabolas by quadratic formula

Sketch $y = -x^2 + 6x + 1$

1. Concavity: $a < 0$, concave down.

2. y-int: Let $x = 0$, $y = 1$

3. x-int:

$$x = \frac{-6 \pm \sqrt{6^2 - 4(-1)(1)}}{2(-1)}$$

$$= \frac{-6 \pm \sqrt{40}}{-2}$$

$$= \frac{-6 \pm 2\sqrt{10}}{-2}$$

$$= 3 \pm \sqrt{10}$$

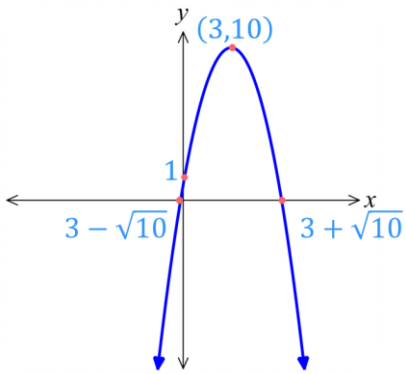
4. Vertex:

$$x = \frac{-6}{2(-1)} = 3$$

$$y = -(3)^2 + 6(3) + 1$$

$$= 10$$

Vertex at $(3,10)$



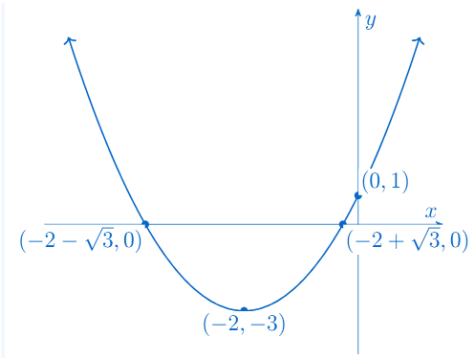
Sketch $y = x^2 + 4x + 1$

1. Concavity:

2. y-int:

3. x-int:

4. Vertex:



Guided Practice

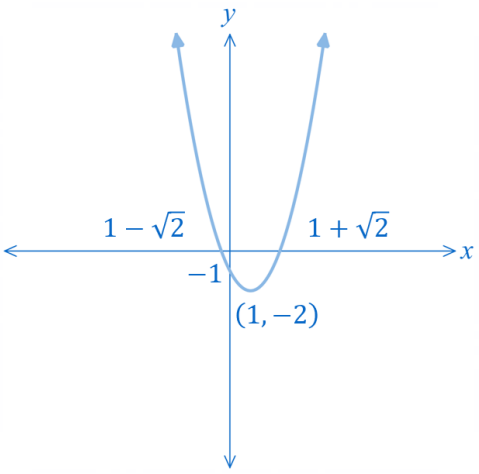
a $y = x^2 - 2x - 1$

1. Concavity:

2. y-int:

3. x-int:

4. Vertex:



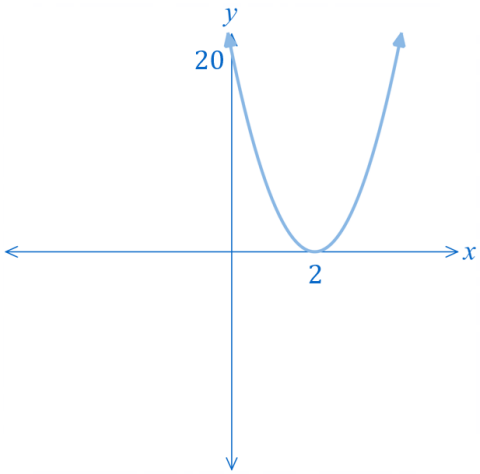
b $y = 5x^2 - 20x + 20$

1. Concavity:

2. y-int:

3. x-int:

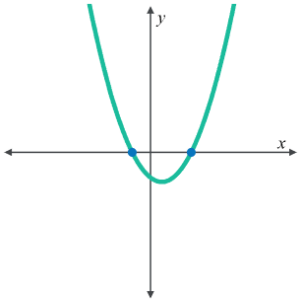
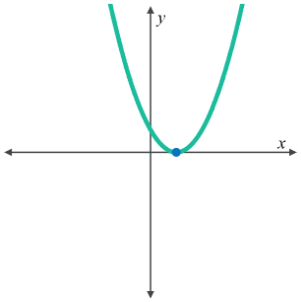
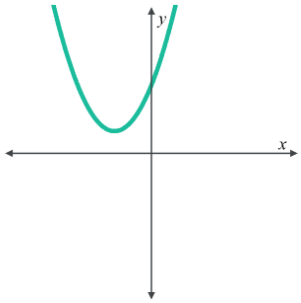
4. Vertex:



ⓘ The Discriminant and the Graph

The **discriminant** $\Delta = b^2 - 4ac$ tells **how many solutions**, or **x-intercepts**, a quadratic has.

Check the discriminant first; it tells you if x-intercepts exist.

$\Delta > 0$	$\Delta = 0$	$\Delta < 0$
2 solutions	1 solution	No real solutions
		
	We say the parabola is **tangent** to the x-axis.	

Example Sketch parabolas by checking the discriminant first

Sketch $y = 3x^2 - 6x + 4$

1. Concavity: $a > 0$, concave up.
2. y-int: Let $x = 0$, $y = 4$
3. x-int:

$\Delta = b^2 - 4ac$

$= (-6)^2 - 4(3)(4)$

$= -12$

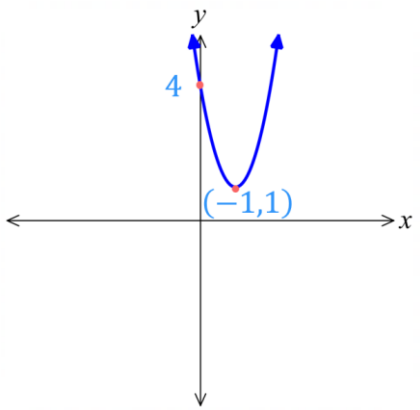
No x intercepts.

4. Vertex:

$x = \frac{-b}{2a} = \frac{-(-6)}{2(3)} = 1$

$y = 3(1)^2 - 6(1) + 4 = 1$

Vertex at $(1,1)$



How did we know there were no x-intercepts?

The value of the discriminant was ……

What would have happened if you tried to use the quadratic formula?

You would be square rooting a ……

How did finding the discriminant save us time?

We did not need to attempt to find the …… intercepts.

How did we find the vertex?

We found the x-coordinate using the formula ……., then substituted into the …… to find y.

Guided Practice

Sketch these parabolas by checking the discriminant first.

$y = x^2 - 4x + 5$

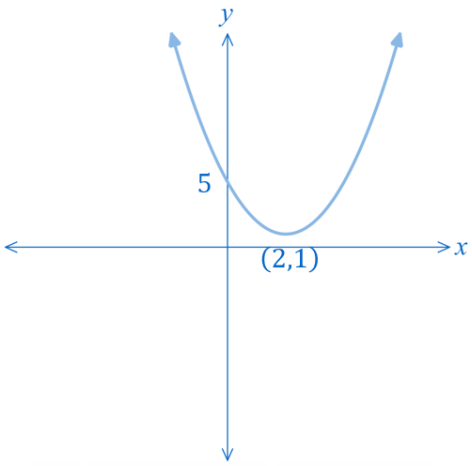
1. Concavity:

2. y-int:

3. x-int:

$\Delta = b^2 - 4ac$

4. Vertex:



$y = -4x^2 - 13x - 53$

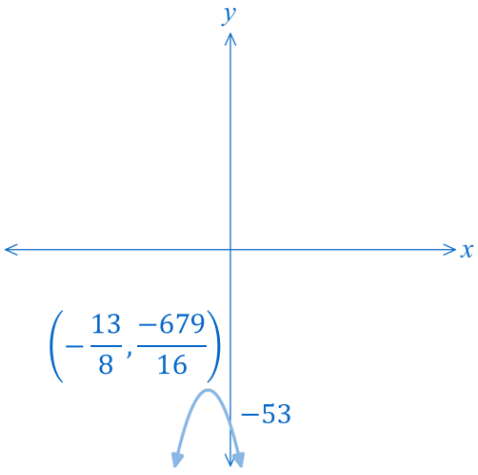
1. Concavity:

2. y-int:

3. x-int:

$\Delta = b^2 - 4ac$

4. Vertex:

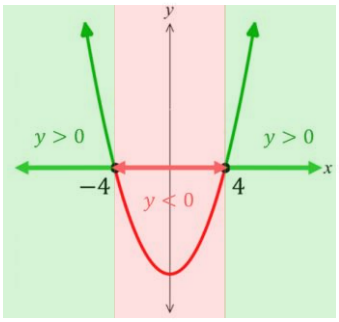


Finding the Equation of a Parabola from its Graph

Quadratic Inequalities

Identify the region where $y > 0$ or $y < 0$

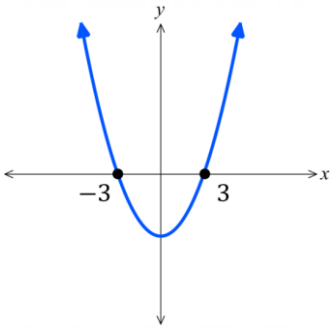
a



$y > 0$: $x < -4,\ x > 4$

$y < 0$: $-4 < x < 4$

b



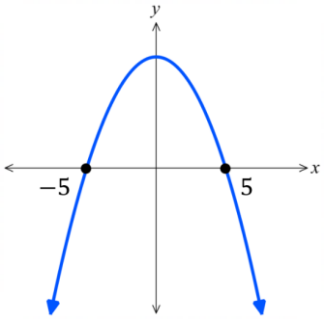
$y > 0$:

$x < -3,\ x > 3$

$y < 0$:

$-3 < x < 3$

c



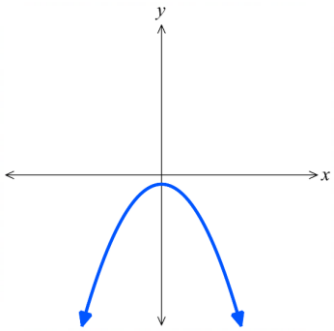
$y > 0$:

$-5 < x < 5$

$y < 0$:

$x < -5,\ x > 5$

d



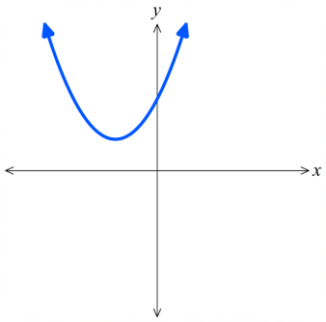
$y > 0$:

no solution

$y < 0$:

all real x

e



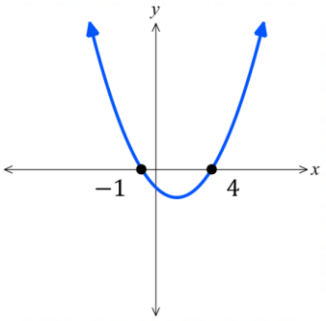
$y > 0$:

all real x

$y < 0$:

no solution

f



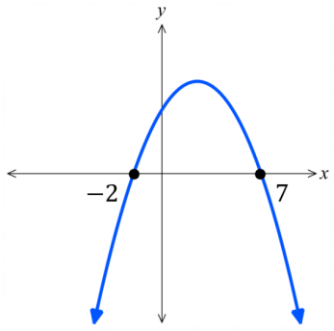
$y > 0$:

$x < -1,\ x > 4$

$y < 0$:

$-1 < x < 4$

g



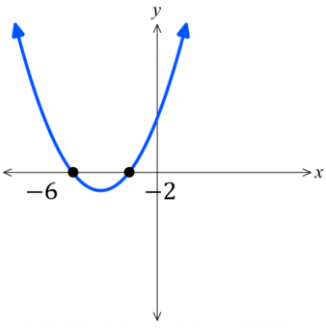
$y > 0$:

$-2 < x < 7$

$y < 0$:

$x < -2,\ x > 7$

h



$y > 0$:

$x < -6,\ x > -2$

$y < 0$:

$-6 < x < -2$

Quadratic Inequalities

A quadratic inequality is an inequality in which the highest power of the pronumeral is 2.
Solving quadratic inequalities is best done by sketching a parabola.

1. Move all terms to one side so that you have $ax^2 + bx + c \geq 0$ or ≤ 0.
2. Sketch the parabola $y = ax^2 + bx + c$, **showing all x-intercepts**.
3. Examine where the graph is above or below the x-axis.

Example Solve quadratic inequalities using graphing techniques

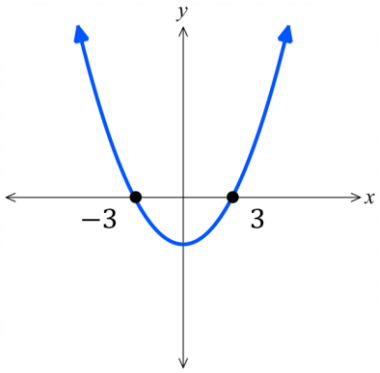
Solve $x^2 > 9$

1.

$$x^2 - 9 > 0$$

2.

$$(x + 3)(x - 3) > 0$$



3. $x < -3$ or $x > 3$

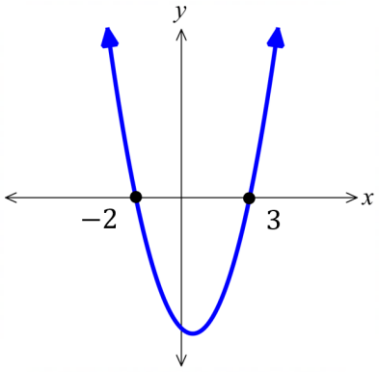
Solve $x + 6 \geq x^2$

1.

$$-x^2 + x + 6 \geq 0$$
$$x^2 - x - 6 \leq 0$$

2.

$$(x - 3)(x + 2) \leq 0$$



3. $-2 \leq x \leq 3$

Guided Practice

a Solve $x^2 + 2x > 0$

$x < -2,\ x > 0$

b Solve $x^2 - x - 12 \leq 0$

$-3 \leq x \leq 4$

Points of Intersection of a Parabola and a Line

ⓘ Finding the points of Intersection of a Parabola and a Line

When you have two graphs: $y = ax^2 + bx + c$ and $y = mx + d$. The **point of intersection** lies on both curves, so is the **simultaneous solution** to the equations.

1. Set the two y values to be equal $y_1 = y_2$
2. Solve for x by rearranging the resulting quadratic equation.
3. Solve for y.

Example Find points of intersection of a parabola and a line

Find the points of intersection of $y = x^2 + 4x + 4$ and $y = 16$.

1. $x^2 + 4x + 4 = 16$
2. $x^2 + 4x - 12 = 0$

$(x + 6)(x - 2) = 0$

$x = -6$ or 2

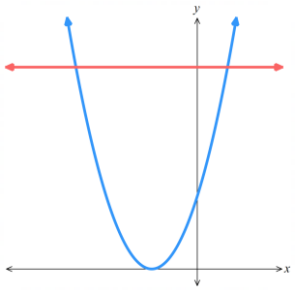
3. when $x = -6$, $y = 16$
   when $x = 2$, $y = 16$

so, points of intersection are $(-6,16)$ and $(2,16)$

Guided Practice

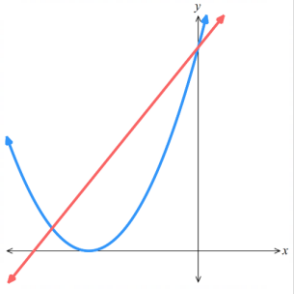
Find the points of intersection of:

a $y = x^2 + 6x + 9$ and $y = 25$



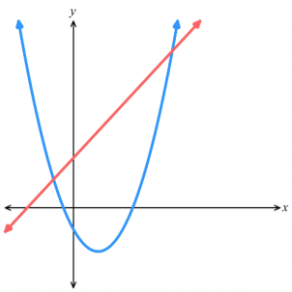
$(-8,25)$ and $(2,25)$

b $y = x^2 + 6x + 9$ and $y = 2x + 9$



$(-4,1)$ and $(0,9)$

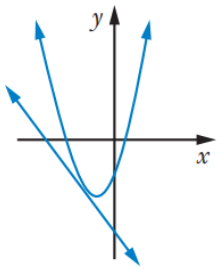
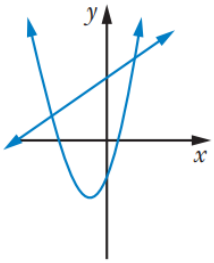
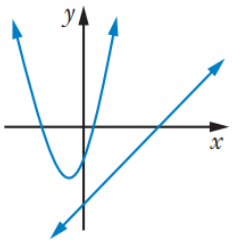
c $y = 2x^2 - 5x - 3$ and $y = 3x + 7$



$(-1,4)$ and $(5,22)$

ⓘ The Discriminant and Points of Intersection

A parabola intersects a straight line at **zero**, **one**, or **two** points.
The discriminant of the resulting equation after setting the two equations equal reveals the number of points of intersection.

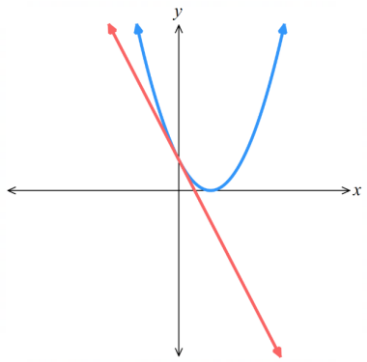
$\Delta = 0$	$\Delta > 0$	$\Delta < 0$
1 intersection point	Two intersection points	No intersection points
		
the line is **tangent** to the parabola		

🔍 **Investigation** The discriminant and points of intersection

Consider the two graphs of $y = x^2 - 2x + 1$ and $y = -2x + 1$.

The graph shows that there is only one point of intersection.

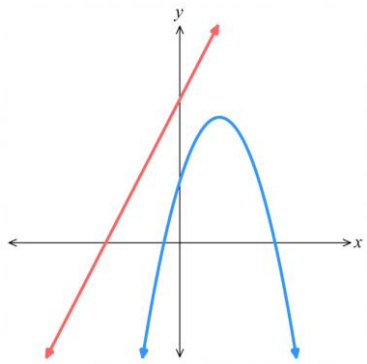
Show that $\Delta = 0$ and then find the point of intersection.



Consider the two graphs of $y = -2x^2 + 5x + 3$ and $y = 3x + 7$.

The graph shows no points of intersection.

Find the value of Δ and confirm this is true.



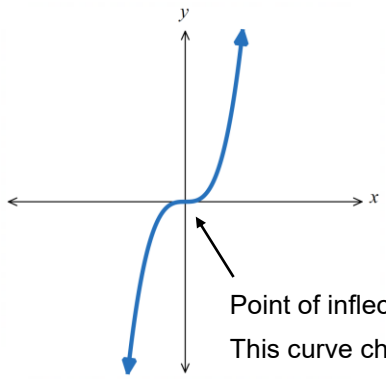
Cubic Curves and Transformations

ⓘ Cubic Curves

A cubic function has an equation where x has a highest power of 3.

The most basic form is $y = kx^3$

The basic cubic curve has a **point of inflection**, where the **concavity changes**.



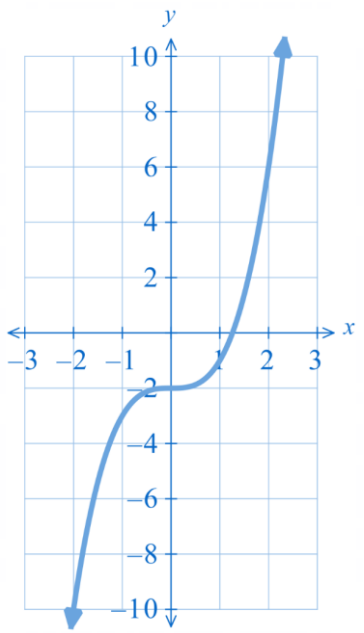
Point of inflection

This curve changes from concave down to concave up

🔍 **Investigation** Graph cubic relationships using a table of values

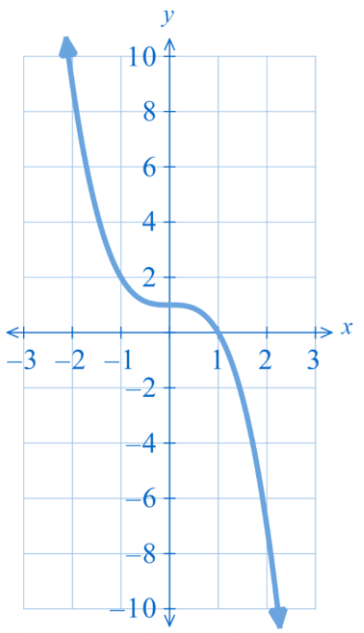
Draw the graph of $y = x^3 - 2$

x	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2
y							



Draw the graph of $y = -x^3 + 1$

x	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2
y							



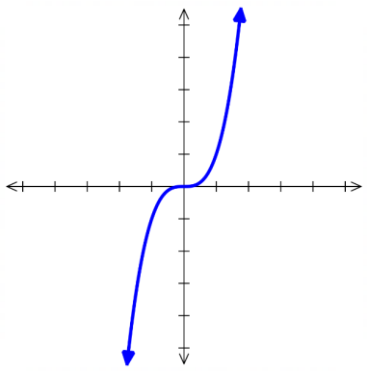
ⓘ Cubic Curve Transformations

For a cubic curve $y = k(x - b)^3 + c$

The parameters b, c, k affect the graph in predictable ways.

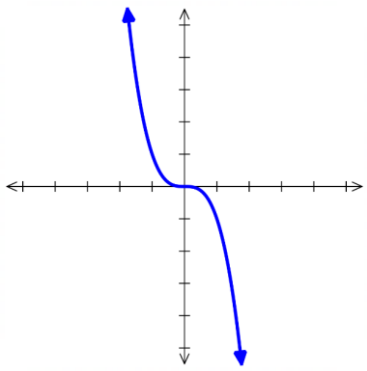
$k > 0$, the curve is increasing.

$y = 1x^3$



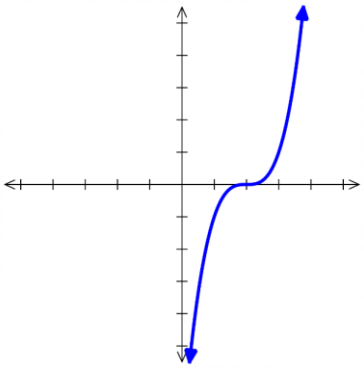
$k < 0$, the curve is decreasing

$y = -1x^3$



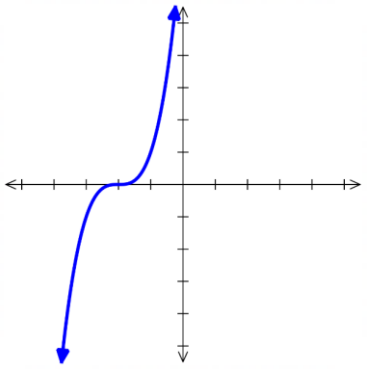
$b > 0$, the curve is shifted ……

$y = (x - 2)^3$



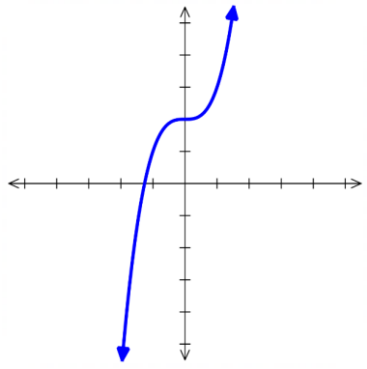
$b < 0$, the curve is shifted ……

$y = (x + 2)^3$



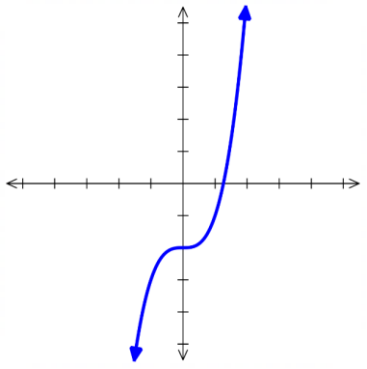
$c > 0$, the curve is shifted ……

$y = x^3 + 2$



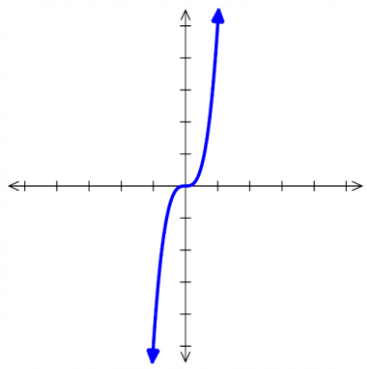
$c < 0$, the curve is shifted ……

$y = x^3 - 2$



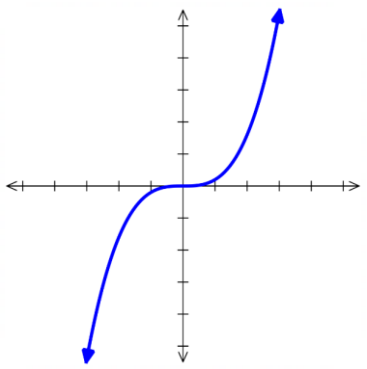
$|k| > 1$, the curve gets narrower.

$y = 5x^3$



$0 < |k| < 1$ the curve gets wider.

$y = \frac{1}{5}x^3$



Inflection Point Form of a Cubic Curve

The **inflection point form** of a cubic reveals its **inflection point** because it shows the graph's horizontal and vertical shifts.

$$y = a(x-h)^3 + k$$

Concavity — inflection point (h,k)

To sketch a cubic curve:

1. Determine the behaviour of curve by substituting $x \to \infty$.
2. Find y-intercept by letting $x = 0$.
3. Find x-intercepts by letting $y = 0$.
4. Identify the point of inflection (h,k).

Example Success Criteria

Sketch $y = x^3 - 8$

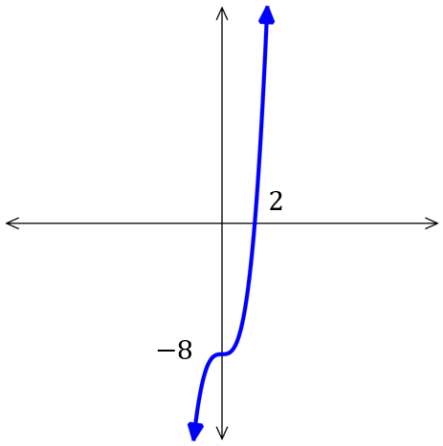
1. Behaviour — When $x \to \infty, y \to \infty$
2. y-int — Let $x = 0$

$$y = 0^3 - 8$$
$$= -8$$

3. x-int — Let $y = 0$

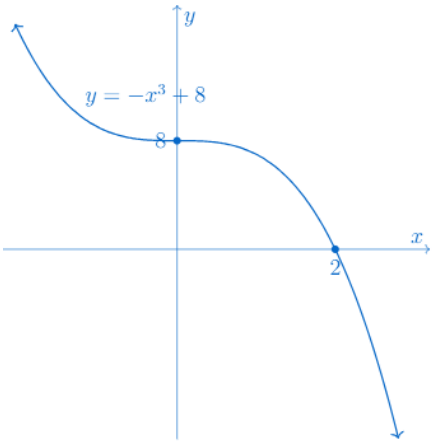
$$0 = x^3 - 8$$
$$8 = x^3$$
$$2 = x$$

4. Point of inflection at $(0,-8)$



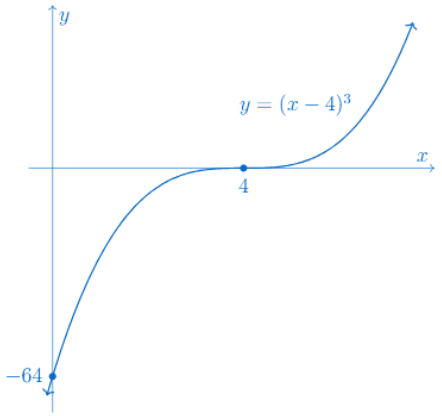
Guided Practice

Sketch $y = -x^3 + 8$

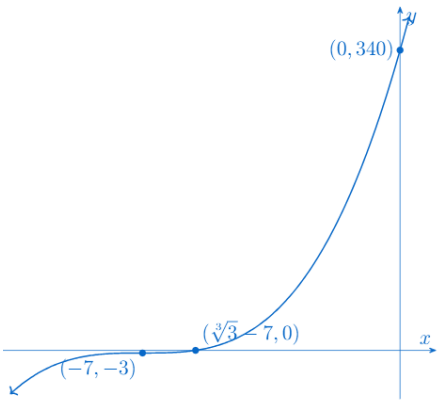


Guided Practice

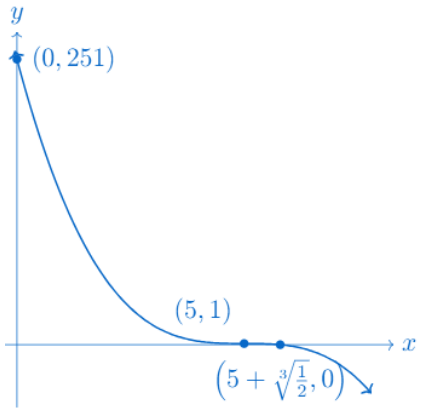
Sketch $y = (x-4)^3$



Sketch $y = (x+7)^3 - 3$



Sketch $y = -2(x-5)^3 + 1$



Power Curve Transformations

ⓘ Power Curve Transformations

For a power curve $y = k(x - b)^n + c$

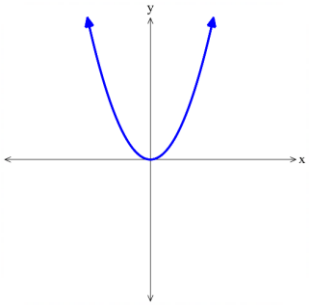
The parameters b, c, k, n affect the graph in predictable ways.

n is even, curve has a turning point.

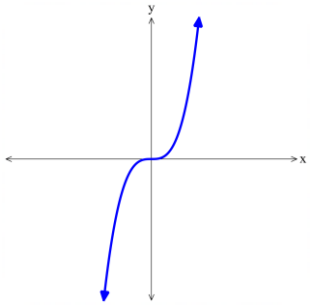
n is odd, curve has inflection point.

As n increases, the curve gets …… but the turning/inflection point gets ……

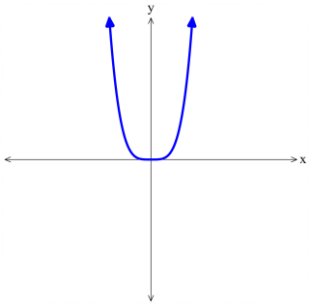
$y = x^2$



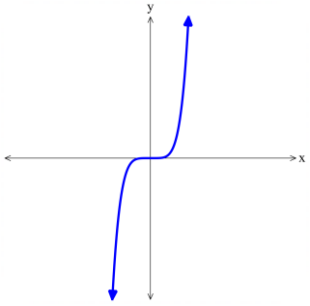
$y = x^3$



$y = x^4$

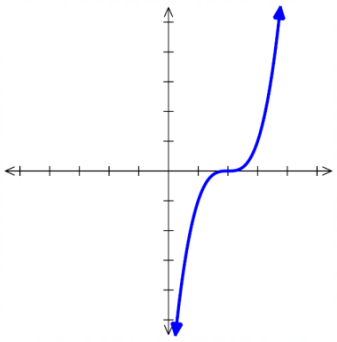


$y = x^5$



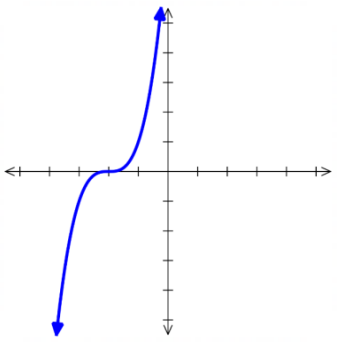
$b > 0$, the curve is shifted ……

$y = (x - 2)^3$



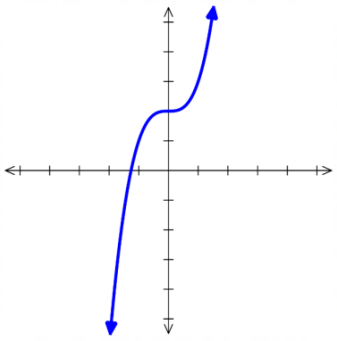
$b < 0$, the curve is shifted ……

$y = (x + 2)^3$



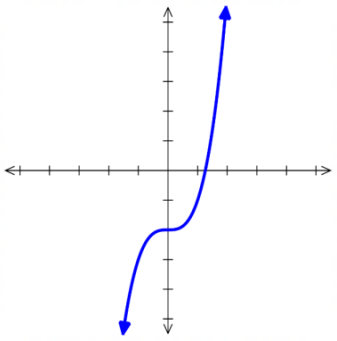
$c > 0$, the curve is shifted ……

$y = x^3 + 2$



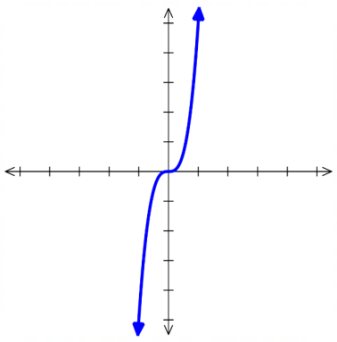
$c < 0$, the curve is shifted ……

$y = x^3 - 2$



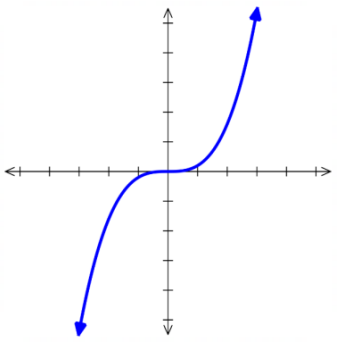
$|k| > 1$, the curve gets narrower.

$y = 5x^3$



$0 < |k| < 1$ the curve gets wider.

$y = \frac{1}{5}x^3$



Sketching Cubics in Factored Form

ⓘ Leading Term of a Cubic

For a cubic equation $y = ax^3 + bx^2 + cx + d$, $\boldsymbol{ax^3}$ is called the **leading term.**

It has the biggest impact on the end behaviour of the curve.

The **constant term**, $\boldsymbol{d}$, will tell us the y-intercept of the curve.

🔍 **Identify** the leading term of a cubic

To find the leading term, multiply the highest power terms from each factor.

To find the constant term, multiply the constant terms of each factor.

$y = (2x - 3)(x - 4)(x + 5)$	$y = (3 - 2x)(x - 4)(x + 5)$
Leading term: $2x^3$	Leading term: $-2x^3$
Constant term: 60	Constant term: -60

Find the leading term and constant term of these cubics.

a $y = (x + 2)(x - 1)(x - 3)$	**b** $y = -(x - 2)(x - 1)(x - 3)$
c $y = x(x + 5)(2x + 3)$	**d** $y = -3x(x - 5)(2x + 15)$
e $y = (3 - x)(2 + x)(x + 5)$	**f** $y = (2x - 3)(3x + 1)(7 - x)$

ⓘ Sketching a Cubic Curve in Factored Form

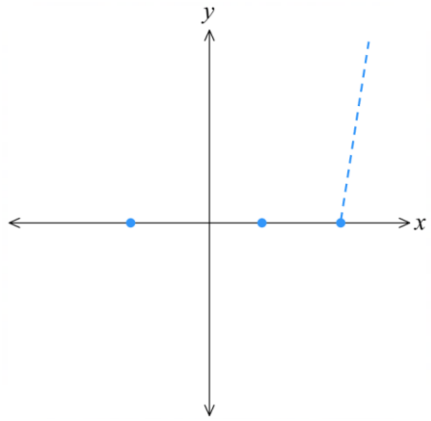
x-intercepts

$$y = k(x-\alpha)(x-\beta)(x-\gamma)$$

1. Find the leading term and determine the end behaviour as $x \to \infty$.
2. Find x intercepts.
3. Find y intercept.

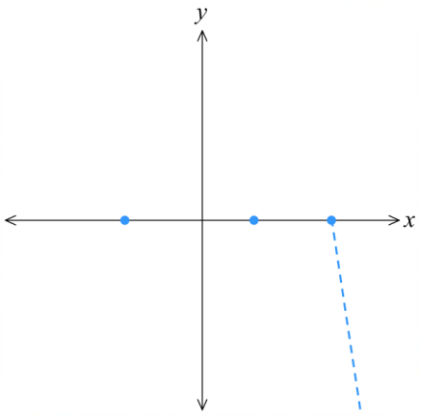
Leading term positive

As $x \to \infty$, $y \to \infty$



Leading term negative

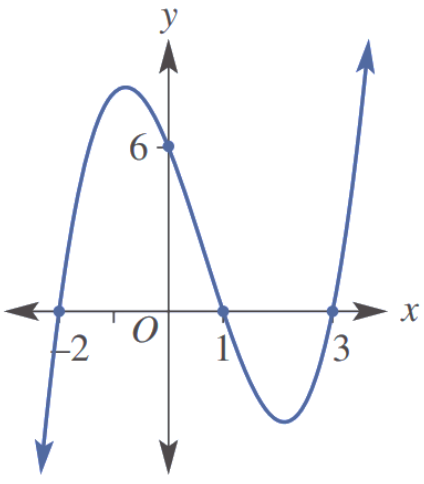
As $x \to \infty$, $y \to -\infty$



Example Sketch cubic curves in factored form

Sketch $y = (x+2)(x-1)(x-3)$

1. The leading term is x^3,
   so when $x \to \infty$, $y \to \infty$
2. Let $y = 0$
   $0 = (x+2)(x-1)(x-3)$
   $x = -2, 1, 3$
3. Let $x = 0$
   $y = (0+2)(0-1)(0-3)$
   $= (2)(-1)(-3)$
   $= 6$



Guided Practice

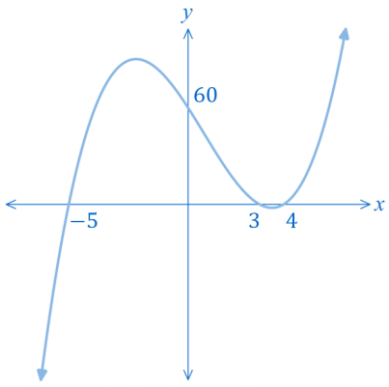
Sketch these cubic curves.

a $y = (x - 4)(x + 5)(x - 3)$

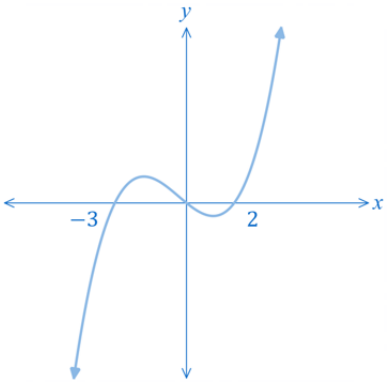
1. leading term & behaviour

2. x-intercepts

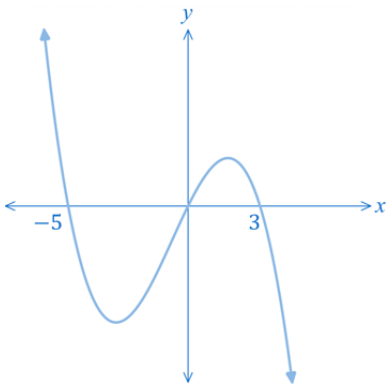
3. y-intercept



b $y = x(x - 2)(x + 3)$



c $y = (x - 4)(x + 5)(x - 3)$



d $y = (3 - x)(x + 1)(2x + 1)$

